LLM-Driven Trading Agent: LeafTrade

> A disciplined, explainable LLM-driven trading system designed for inflation-beating, risk-controlled investing, not speculation, but structured, systematic execution.

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Problem Framing & Design Logic

The Real Problem

Graduate students and disciplined investors don't need lottery-style returns. They need a system that preserves capital, beats inflation over time, and avoids catastrophic drawdown.

The Market Gap

Most retail and algorithmic strategies ignore risk management, overtrade on noise, or rely solely on price data, discarding the rich context embedded in financial news and market events.

Our Thesis

"Trade only when multiple independent signals agree, otherwise hold cash." Inaction is a legitimate position. Capital preservation is not failure.

Why LLMs?

Large Language Models are powerful, but only when applied to tasks they are genuinely suited for. LeafTrade is built around an honest assessment of LLM capabilities.

What LLMs Do Well

- Interpreting unstructured financial text
- Extracting nuanced sentiment from news headlines
- Contextualizing events and market-moving language

What LLMs Do Poorly

- Predicting future price movements
- Performing precise mathematical forecasting
- Acting as a stand-alone decision engine

Our Positioning: "The LLM is not the trader, it is a signal interpreter." This distinction is foundational to every design decision in LeafTrade.

Agent Architecture Overview

LeafTrade is composed of three distinct, modular layers. Each layer has a single, well-defined responsibility, making the system transparent, auditable, and debuggable.

Data Layer

The Body. Ingests market data (OHLCV, moving averages, RSI, ATR) and financial news headlines in real time.

LLM Brain

The Interpreter. Processes raw news text and converts each headline into a structured sentiment score on a 0–10 scale.

Execution Engine

The Rules Engine. Applies deterministic risk filters and places trades only when all conditions across all layers are simultaneously satisfied.

Data Pipeline

Market Data Inputs

- OHLCV (Open, High, Low, Close, Volume) at daily resolution
- 50-day and 200-day moving averages
- RSI (Relative Strength Index) for momentum measurement
- ATR (Average True Range) for volatility sizing

News Pipeline & Deduplication

- Automated financial headline collection across major sources
- **Deduplication via Jaccard Similarity** — identical stories from multiple outlets are collapsed to prevent artificial consensus inflation
- Time decay weighting — older articles carry progressively less signal weight

Why this matters: "We remove duplicate news using Jaccard similarity to avoid fake consensus." Without deduplication, a single story reprinted across 20 outlets would masquerade as 20 independent signals, a critical flaw in naive pipelines.

The LLM "Brain"

Model Selection

LeafTrade uses a domain-specific **financial sentiment model**, not a general-purpose GPT. This is deliberate: general models trained on broad corpora introduce noise when applied to tightly scoped financial language.

Output Structure

- Classification: **Positive / Neutral / Negative**
- Confidence score mapped to a normalized **0–10 sentiment scale**
- Aggregated across all non-duplicate headlines for the ticker

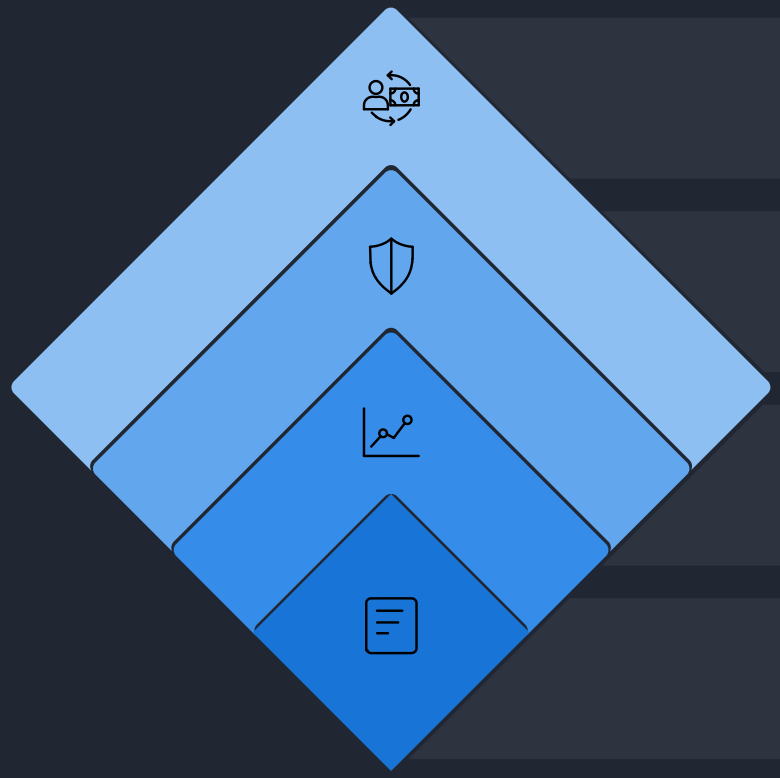
Key Design Decision

The LLM is configured for **deterministic outputs**, temperature is set to zero. This ensures that identical inputs produce identical outputs across backtesting runs, preserving reproducibility and preventing look-ahead contamination.

This is not a minor implementation detail. It is a prerequisite for any intellectually honest backtest.

Decision Flow

Every trade follows the same structured evaluation path. No step can be skipped, and no single strong signal can override a failed condition at any other layer.



Execute Trade
Only if ALL conditions align

Risk Filters
ATR, limits, SPY trend pass

Technical Check
MA50 and RSI confirm signal

News Sentiment
LLM assigns sentiment score

Core Rule: "No trade unless ALL conditions align." This is not a scoring system where partial credit adds up. It is a gate system, one failed condition blocks execution entirely. This design philosophy directly minimizes false positives and impulsive trades driven by a single noisy signal.

Strategy Overview: The 5-Layer System

LeafTrade filters the universe of tradeable opportunities through five sequential, independent quality layers. Each layer is designed to eliminate a distinct category of bad trade.

Long-Term Quality

10-year return history filters out structurally weak equities

Liquidity

Minimum volume thresholds prevent slippage and illiquid fills

Volatility Control

ATR-based filters remove excessively volatile names from consideration

Technical Trend

Price must be in a confirmed uptrend relative to MA50

LLM Sentiment

Aggregated news sentiment must clear threshold before any signal is generated

Entry Logic: The Gate System

All Conditions Must Be Met Simultaneously

Price > MA50

Security must be in a confirmed medium-term uptrend. No bottom-fishing.

RSI: 35–55 Pullback Zone

We buy strength, not euphoria. RSI above 55 signals potential overextension.

Sentiment Score ≥ 6

LLM-derived sentiment must reflect a meaningfully positive news environment.

≥ 70% Consensus & ≥ 5 Headlines

Majority agreement across a minimum number of independent, deduplicated sources.

Design Philosophy

This is a **gate system, not a scoring system.**

In a scoring system, a very strong sentiment score could compensate for a weak technical condition. In LeafTrade, that is explicitly prohibited. **One failure blocks the trade.**

This architecture reflects a core belief: the cost of a missed opportunity is always lower than the cost of a bad trade entered with incomplete conviction.

Risk Controls

Hard Trade-Level Controls

ATR Trailing Stop

3.5× ATR, dynamically widens in volatile markets, tightens as price consolidates

Take Profit

12% hard cap — profit is locked in before greed extends a trade beyond its thesis

Risk Per Trade

Maximum 2% of portfolio at risk on any single position entry

Position Cap

15% maximum allocation to any single security, prevents concentration risk

Market-Wide Protection

Regime Detection: When SPY trades below its 50-day moving average, LeafTrade automatically reduces all position sizes. This systemic risk filter recognizes that individual stock analysis is insufficient during broad market downturns, a principle drawn directly from institutional risk practice.

This macro overlay transforms LeafTrade from a stock-picker into a true portfolio management system, one that distinguishes between security-specific risk and systemic market risk.

Guardrails: Preventing LLM Mistakes

LLMs are powerful signal interpreters, but without guardrails, they are also a source of systematic error. LeafTrade treats the LLM output with healthy skepticism at every stage.

Deduplication

Jaccard similarity removes near-identical articles before the LLM ever sees them. One story published by 15 outlets counts as one data point.

Minimum Article Count

A sentiment reading based on fewer than 5 independent headlines is discarded entirely. Thin coverage cannot generate a trade signal.

Consensus Threshold

At least 70% of scored headlines must agree in sentiment direction before the score is considered valid. Divided opinion = no signal.

Time Decay Weighting

Recent articles carry full weight; older articles decay exponentially. Stale sentiment does not drive current decisions.

Sentiment Momentum Check

Single-period sentiment spikes are filtered. A sustained directional shift is required before sentiment contributes to a gate pass.

"The LLM is treated as one input, never as truth." This design principle, applied consistently, is what separates an academic exercise from a production-grade system.

Validation & Backtesting

Backtest Results

+8.77%

Total Return

0.93

Sharpe Ratio

-2.0%

Max Drawdown

Benchmark: SPY returned **+12.8%** over the same period.

Interpreting the Gap

LeafTrade returned +8.77% against SPY's +12.8%. This 4-point underperformance in a strong bull market is not a system failure, it is the system working exactly as intended.

A Sharpe ratio of 0.93 with a maximum drawdown of only -2.0% indicates that every unit of return was earned with exceptional capital efficiency. LeafTrade never took on the tail risk required to chase benchmark performance.

> **"We underperform in bull markets because we prioritize risk control."** In bear market conditions, this system's drawdown profile would likely invert the comparison entirely.

Limitations & Honest Reflection

Known Constraints

→ Small Stock Universe

LeafTrade was tested on a curated subset of large-cap equities. Performance characteristics may differ materially across mid-cap, small-cap, or international securities with thinner news coverage.

→ Sentiment Lag

LLM sentiment is derived from published headlines, which inherently trail real-time price discovery. In fast-moving events, the system may react one news cycle behind the market.

→ Paper Trading Only

All results are simulated. Live execution introduces slippage, latency, and market impact that backtesting cannot fully replicate.

The Honest Position

These limitations are not embarrassments, they are evidence of a disciplined process. A system that acknowledges its constraints is more trustworthy than one that claims universal applicability.

"This is evidence of a disciplined process, not guaranteed outperformance."

The goal of LeafTrade was never to beat the S&P 500 in every market environment. It was to build a systematic, explainable, risk-controlled framework that a disciplined investor could trust across market cycles.

Live Demo

> Please see our live demonstration of the LeafTrade system in action.

The demo walks through a full end-to-end trade lifecycle: from raw news ingestion and LLM sentiment scoring, through technical signal evaluation and risk gate application, to a final trade decision and simulated execution. Every layer described in this presentation is observable in real time.

Signal Interpretation

Live LLM sentiment scoring on current financial headlines

Gate Evaluation

Real-time pass/fail status across all five decision layers

Trade Decision Log

Fully explainable output — every decision is logged with its rationale